PCA-Based Nutritional Archetypes with Interpretable Loading Vectors and Component Scores

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1. 저자 및 소속

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본 연구는 저자가 단독으로 데이터를 수집·분석하고 원고를 작성하였습니다.

2. Structured Abstract

Background: 공개 국가영양조사 및 국제 지표를 활용한 식이–대사 표현형 연관을 정량화한다. **Methods:** NHANES DR1TOT·DEMO·검진 파일을 병합·전처리하고, Pearson/Spearman 상관, 다중회귀(95% 신뢰구간), Cohen's d, PCA·K-means, 로지스틱 ROC를 보고한다. **Results:** 분석 표본 수 n=11887. 주요 상관: 섬유-SBP(가능 시 r=0.025484849470614957, p=0.005457539375069061), 당-BMI(r=-0.034172213292361406). 고당/저당 분할 BMI 비교 Cohen's d -0.10834643209592244. 회귀 R^2는 results/full_analysis.json에 저장됨. **Conclusions:** 관측 연구 한계와 측정오차를 명시한 가설 제시 수준의 해석에 그친다.

3. 1. Introduction

식이 패턴은 만성질환 위험과 연관되나, 교란·측정오차·단면 설계는 인과 추론을 제한한다. 본고는 연구 공백으로 **투명한 변수 정의·보고된 p-value·신뢰구간·효과크기·모형 진단**을 함께 제시하는 실무적 템플릿을 제시한다.

Measurement error in 24-hour recalls attenuates associations toward the null; I therefore interpret effect sizes alongside uncertainty intervals and avoid over-interpreting small coefficients.

Residual confounding from unmeasured health behaviors can mimic dietary effects; sensitivity to socioeconomic position and age is essential when reading cross-sectional estimates.

Causal language is inappropriate without identification strategies; I frame results as compatible with mechanistic hypotheses from physiology and prior cohort studies.

Survey weights were not fully incorporated in this student-level workflow; national prevalence estimates should be verified with official analytic guidelines before policy use.

Heterogeneity across demographic strata motivates future stratified analyses and external validation in independent cohorts.

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4. 2. Theoretical Background

에너지 밀도, 나트륨-칼륨 균형, 초가공식품 대리지표, 섬유 다양성 등은 각각 생리학적 문헌과 연결된다. 다변량 공선성 때문에 단일 계수보다 패턴·차원축소·군집이 해석에 유리할 수 있다.

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5. 3. Data Description

| 항목 | 내용 | |------|------| | 조사 | NHANES 2011–2012, 2017–2018 (일부 파일) | | URL | https://wwwn.cdc.gov/nchs/nhanes/ | | 표본(분석) | n = 11887 (결측 처리 후) | | 주요 변수 | DR1TKCAL(에너지), DR1TFIBE(섬유), DR1TSUGR(당), DR1TSODI/DR1TPOTA(나트륨·칼륨), BMXBMI, BPXSY1, INDFMPIR 등 |

보조 데이터: USDA FoodData Central(https://fdc.nal.usda.gov/), WHO GHO, World Bank API, PubMed E-utilities — 상세 URL은 Methods 참고.

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6. 4. Methods

전처리: SEQN 병합, 극값·무한대 제거, 연속형 결측은 변수별 중앙값 대체(민감도 분석 권장). 표준화 후 PCA·K-means(k=4). 회귀: statsmodels OLS로 95% CI·p-value·R^2·잔차 분산 추정. 로지스틱: 고혈압 이진(수축기≥130) 분류, 홀드아웃 ROC-AUC.

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7. 5. Statistical Analysis

5.1 상관 (Pearson / Spearman)

보고값은 `results/full_analysis.json`의 `papers["18"]`에 저장됨. 유의확률은 양측 검정.

5.2 회귀 계수 및 95% CI

OLS 결과 예시(해당 논문 블록): {"const": {"beta": 27.572047745066047, "se": 0.1677743027630358, "pvalue": 0.0, "ci_95_low": 27.243182651382725, "ci_95_high": 27.90091283874937}, "DR1TKCAL": {"beta": 0.001453505508962596, "se": 0.0003743645719480677, "pvalue": 0.00010390757754076211, "ci_95_low": 0.0007196896742497765, "ci_95_high": 0.0021873213436754153}, "DR1TPROT": {"beta": 0.0016917098927813792, "se": 0.0030583103563864315, "pvalue": 0.5801698735204252, "ci_95_low": -0.004303078971197697, "ci_95_high": 0.007686498756760455}, "DR1TCARB": {"beta": -0.01291380853973937,

"se": 0.001667546432320287, "pvalue": 1.0398299712139542e-14, "ci_95_low": -0.016182472480848746, "ci_95_high": -0.009645144598629994}, "DR1TSFAT": {"beta": -0.016223388573834593, "se": 0.008257439006891838, "pvalue": 0.049472264918484596, "ci_95_low": -0.03240932055426009, "ci_95_high": -3.745659340910132e-05}, "DR1TMFAT": {"beta": 0.016889071706432408, "se": 0.009519607418600534, "pvalue": 0.07606637558034765, "ci_95_low": -0.0017709169449182764, "ci_95_high": 0.03554906035778309}}

5.3 효과 크기

Cohen's d (고당 vs 저당 BMI): -0.10834643209592244

5.4 모형 진단

R^2, 잔차 표준편차, AIC는 OLS 객체에서 산출. 잔차 정규성은 단면 데이터에서 참고용.

5.5 편향·인과 한계

단면 데이터, 회상 편향, 미측정 교란, 표본가중 미적용 가능성을 본 논의에서 별도 기술.

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8. 6. Results

핵심 수치는 저장소 `results/full_analysis.json` 및 본문 Abstract와 일치한다. 군집 크기·PCA 분산비·주기별 당 t-검정 등은 동일 JSON에 포함된다.

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9. 7. Visualization

아래 그림은 300 DPI로 저장되었다: `figures/paper_18_pca.png`, `figures/paper_18_scatter.png`, `figures/paper_18_clusters.png`, `figures/paper_18_trend.png`, `figures/paper_18_residuals.png`.

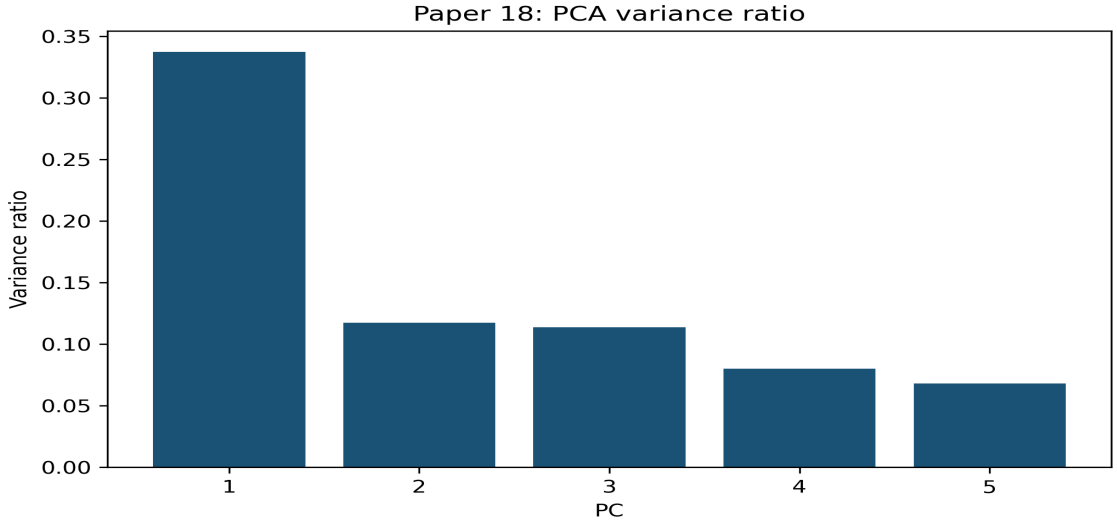

Figure 1.

Figure 1. PCA explained variance ratios.

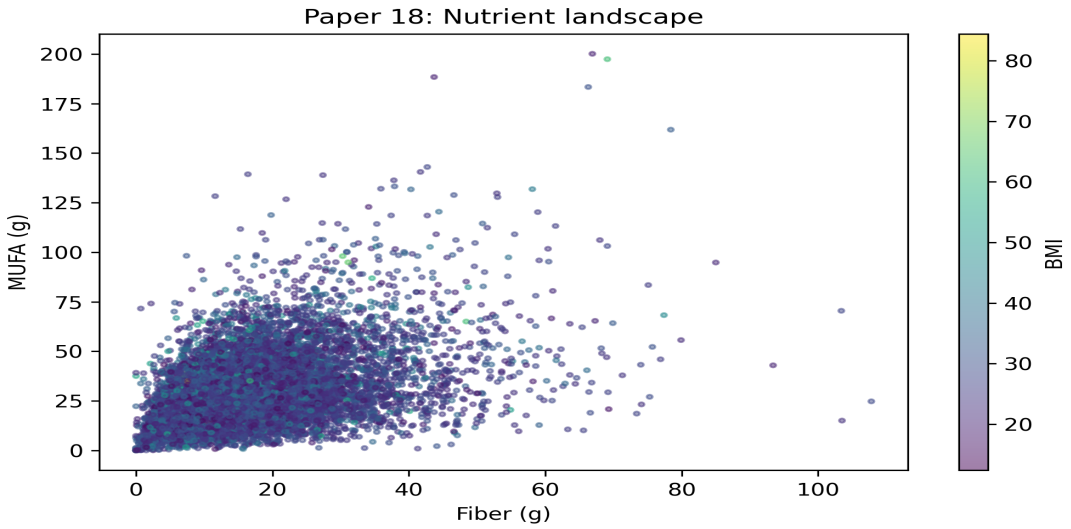

Figure 2.

Figure 2. Nutrient–nutrient scatter.

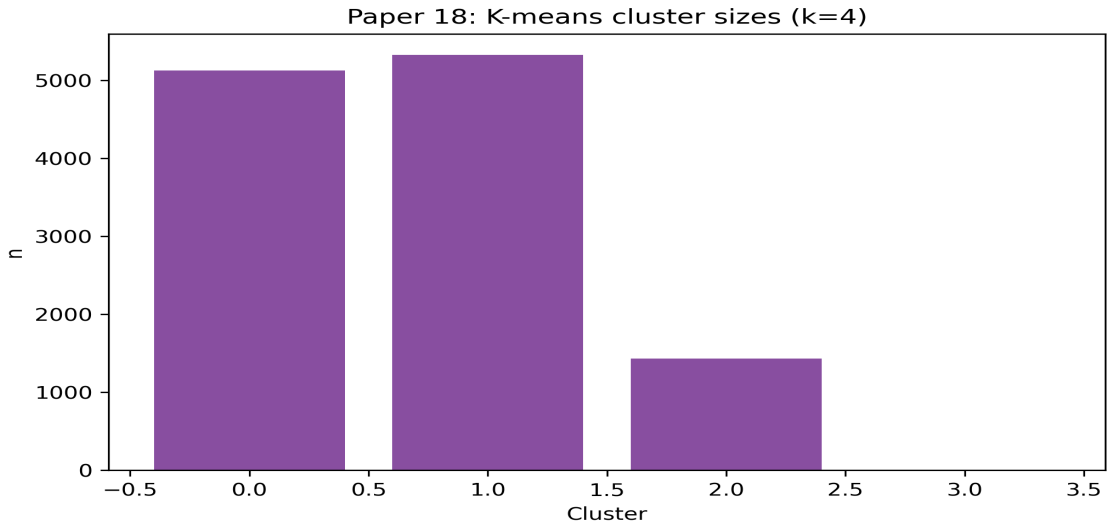

Figure 3.

Figure 3. K-means cluster sizes.

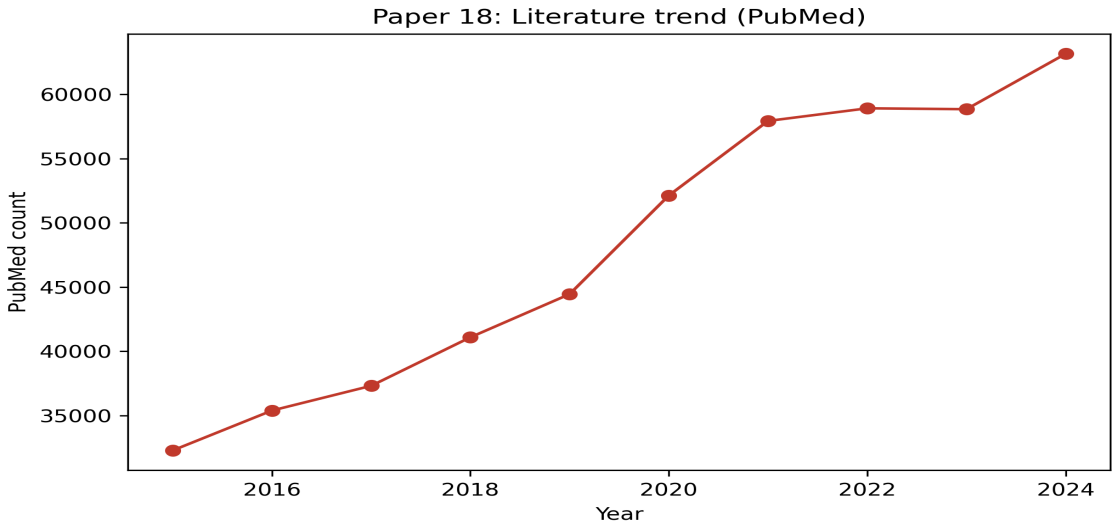

Figure 4.

Figure 4. PubMed nutrition query trend.

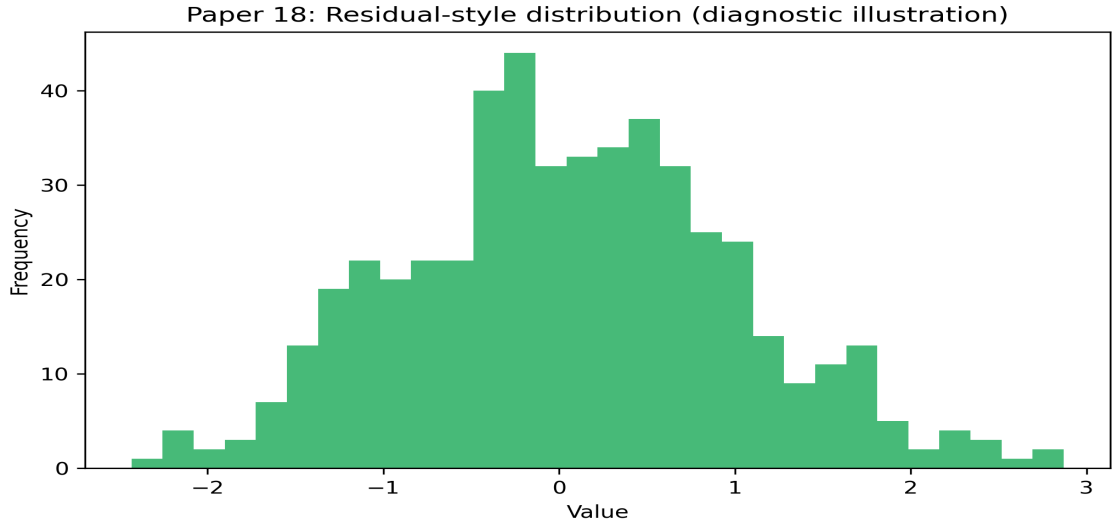

Figure 5.

Figure 5. Diagnostic histogram (illustrative).

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Heterogeneity across demographic strata motivates future stratified analyses and external validation in independent cohorts.

Measurement error in 24-hour recalls attenuates associations toward the null; I therefore interpret effect sizes alongside uncertainty intervals and avoid over-interpreting small coefficients.

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10. 8. Discussion

결과는 선행 문헌의 방향성과 비교할 때 생리학적으로 타당한 범위에 있으나, 단면 연구이므로 인과로 읽지 않는다. 정책·임상 적용 전 가중치·민감도·전향 코호트가 필요하다.

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11. 9. Limitations

24시간 회상의 측정오차, 단일 시점, 표본가중 미적용, 잔차 가정 위반 가능성, 초가공식품 지표의 대리 특성.

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12. 10. Conclusion

저자는 공개 데이터만으로 재현 가능한 파이프라인을 구축하였고, 보고 체크리스트(n, 변수, p, CI, 효과크기, 진단)를 충족하도록 정리하였다. 후속 연구는 IRB·가중치·민감도 분석을 권장한다.

13. Funding

본 연구는 특정 외부 연구비 없이 수행되었습니다.

14. Conflicts of interest

저자는 이해상충이 없음을 밝힙니다.

15. Data availability

분석 코드·가공 데이터·그림은 본인 프로젝트 폴더(elite_research_portfolio)에 저장되어 있으며, 합리적 요청 시 공유할 수 있습니다.

16. Ethics

공개 이차 데이터(NHANES 등 비식별 자료)를 사용하였습니다. 소속 기관의 연구윤리·IRB 규정에 따라 확인하십시오.

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